

INTRODUCTION Partial differential equations (PDEs), like ordinary differential equations (ODEs), are classified as *linear* or *nonlinear*. Analogous to a linear ODE (see (6) of Section 1.1), the dependent variable and its partial derivatives appear only to the first power in a linear PDE. In this and the chapters that follow, we are concerned only with linear partial differential equations.

Linear Partial Differential Equation If we let u denote the dependent variable and x and y the independent variables, then the general form of a **linear second-order partial differential equation** is given by

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G, \quad (1)$$

where the coefficients $A, B, C, \dots, G$ are constants or functions of x and y . When $G(x, y) = 0$, equation (1) is said to be **homogeneous**; otherwise, it is **nonhomogeneous**.

EXAMPLE 1 Linear Second-Order PDEs

The equations

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = xy$$

are examples of linear second-order PDEs. The first equation is homogeneous and the second is nonhomogeneous. $\equiv$

Solution of a PDE A **solution** of a linear partial differential equation (1) is a function $u(x, y)$ of two independent variables that possesses all partial derivatives occurring in the equation and that satisfies the equation in some region of the xy -plane.

It is not our intention to examine procedures for finding *general solutions* of linear partial differential equations. Not only is it often difficult to obtain a general solution of a linear second-order PDE, but a general solution is usually not all that useful in applications. Thus our focus throughout will be on finding *particular solutions* of some of the important linear PDEs, that is, equations that appear in many applications.

We are interested only in particular solutions of PDEs.

Separation of Variables Although there are several methods that can be tried to find particular solutions of a linear PDE, the one we are interested in at the moment is called the method of **separation of variables**. In this method if we are seeking a particular solution of, say, a linear second-order PDE in which the independent variables are x and y , then we seek to find a particular solution in the form of *product* of a function x and a function of y :

$$u(x, y) = X(x)Y(y).$$

With this assumption, it is *sometimes* possible to reduce a linear PDE in two variables to two ODEs. To this end we observe that

$$\frac{\partial u}{\partial x} = X'Y, \quad \frac{\partial u}{\partial y} = XY', \quad \frac{\partial^2 u}{\partial x^2} = X''Y, \quad \frac{\partial^2 u}{\partial y^2} = XY'',$$

where the primes denote ordinary differentiation.

EXAMPLE 2 Using Separation of Variables

Find product solutions of $\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}$.

SOLUTION Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields

$$X''Y = 4XY'.$$

After dividing both sides by $4XY$, we have separated the variables:

$$\frac{X''}{4X} = \frac{Y'}{Y}.$$

Since the left-hand side of the last equation is independent of y and is equal to the right-hand side, which is independent of x , we conclude that both sides of the equation are independent of x and y . In other words, each side of the equation must be a constant. As a practical matter it is convenient to write this real **separation constant** as $-\lambda$. From the two equalities,

$$\frac{X''}{4X} = \frac{Y'}{Y} = -\lambda$$

we obtain the two linear ordinary differential equations

$$X'' + 4\lambda X = 0 \quad \text{and} \quad Y' + \lambda Y = 0. \quad (2)$$

See Example 2, Section 3.9
and Example 1, Section 12.5.

For the three cases for λ : zero, negative, or positive; that is, $\lambda = 0$, $\lambda = -\alpha^2 < 0$, and $\lambda = \alpha^2 > 0$, where $\alpha > 0$, the ODEs in (2) are, in turn,

$$X'' = 0 \quad \text{and} \quad Y' = 0, \quad (3)$$

$$X'' - 4\alpha^2 X = 0 \quad \text{and} \quad Y' - \alpha^2 Y = 0, \quad (4)$$

$$X'' + 4\alpha^2 X = 0 \quad \text{and} \quad Y' + \alpha^2 Y = 0. \quad (5)$$

Case I ($\lambda = 0$): The DEs in (3) can be solved by integration. The solutions are $X = c_1 + c_2 x$ and $Y = c_3$. Thus a particular product solution of the given PDE is

$$u = XY = (c_1 + c_2 x)c_3 = A_1 + B_1 x, \quad (6)$$

where we have replaced $c_1 c_3$ and $c_2 c_3$ by A_1 and B_1 , respectively.

Case II ($\lambda = -\alpha^2$): The general solutions of the DEs in (4) are

$$X = c_4 \cosh 2\alpha x + c_5 \sinh 2\alpha x \quad \text{and} \quad Y = c_6 e^{\alpha^2 y},$$

respectively. Thus, another particular product solution of the PDE is

$$u = XY = (c_4 \cosh 2\alpha x + c_5 \sinh 2\alpha x)c_6 e^{\alpha^2 y}$$

$$\text{or} \quad u = A_2 e^{\alpha^2 y} \cosh 2\alpha x + B_2 e^{\alpha^2 y} \sinh 2\alpha x, \quad (7)$$

where $A_2 = c_4 c_6$ and $B_2 = c_5 c_6$.

Case III ($\lambda = \alpha^2$): Finally, the general solutions of the DEs in (5) are

$$X = c_7 \cos 2\alpha x + c_8 \sin 2\alpha x \quad \text{and} \quad Y = c_9 e^{-\alpha^2 y},$$

respectively. These results give yet another particular solution

$$u = A_3 e^{-\alpha^2 y} \cos 2\alpha x + B_3 e^{-\alpha^2 y} \sin 2\alpha x, \quad (8)$$

where $A_3 = c_7 c_9$ and $B_3 = c_8 c_9$. ≡

It is left as an exercise to verify that (6), (7), and (8) satisfy the given partial differential equation $u_{xx} = 4u_y$. See Problem 29 in Exercises 13.1.

Separation of variables is not a general method for finding particular solutions; some linear partial differential equations are simply not separable. You should verify that the assumption $u = XY$ does not lead to a solution for $\partial^2 u / \partial x^2 - \partial u / \partial y = x$.

Superposition Principle The following theorem is analogous to Theorem 3.1.2 and is known as the **superposition principle**.

Theorem 13.1.1 Superposition Principle

If $u_1, u_2, \dots, u_k$ are solutions of a homogeneous linear partial differential equation, then the linear combination

$$u = c_1 u_1 + c_2 u_2 + \cdots + c_k u_k,$$

where the $c_i, i = 1, 2, \dots, k$ are constants, is also a solution.

Throughout the remainder of the chapter we shall assume that whenever we have an infinite set $u_1, u_2, u_3, \dots$ of solutions of a homogeneous linear equation, we can construct yet another solution u by forming the infinite series

$$u = \sum_{k=1}^{\infty} c_k u_k,$$

where the $c_k, k = 1, 2, \dots$, are constants.

Classification of Equations A linear second-order partial differential equation in two independent variables with constant coefficients can be classified as one of three types. This classification depends only on the coefficients of the second-order derivatives. Of course, we assume that at least one of the coefficients A, B , and C is not zero.

Definition 13.1.1 Classification of Equations

The linear second-order partial differential equation

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G,$$

where A, B, C, D, E, F , and G are real constants, is said to be

hyperbolic if $B^2 - 4AC > 0$,

parabolic if $B^2 - 4AC = 0$,

elliptic if $B^2 - 4AC < 0$.

EXAMPLE 3 Classifying Linear Second-Order PDEs

Classify the following equations:

$$(a) \quad 3 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} \quad (b) \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial y^2} \quad (c) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

SOLUTION (a) By rewriting the given equation as

$$3 \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = 0$$

we can make the identifications $A = 3, B = 0$, and $C = 0$. Since $B^2 - 4AC = 0$, the equation is **parabolic**.

(b) By rewriting the equation as

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0,$$

we see that $A = 1, B = 0, C = -1$, and $B^2 - 4AC = -4(1)(-1) > 0$. The equation is **hyperbolic**.

(c) With $A = 1, B = 0, C = 1$, and $B^2 - 4AC = -4(1)(1) < 0$, the equation is **elliptic**. $\equiv$

REMARKS

(i) Separation of variables is not a general method for finding particular solutions of linear partial differential equations. Some equations are simply not separable. You are encouraged to verify that the assumption $u(x, y) = X(x)Y(y)$ does not lead to a solution of the linear second-order PDE $\partial^2 u / \partial x^2 + \partial u / \partial x = y$.

(ii) A detailed explanation of why the classifications given in Definition 13.1.1 are important is beyond the scope of this text. But you should at least be aware that these classifications do have a practical importance. Beginning in Section 13.3 we are going to solve some PDEs subject to both boundary and initial conditions. The kinds of side conditions appropriate for a given equation depend on whether the equation is hyperbolic, parabolic, or elliptic. Also, we shall see in Chapter 16 that numerical solution methods for linear second-order PDEs differ in conformity with the classification of the equation.